

AI Use Declaration Form

Course: Introduction to Artificial Intelligence

Lab Title: Lab 1 Part A — Uninformed Search Algorithms

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GitHub Repository Link: <https://github.com/kgyamebi/lab-3.git>

Date Submitted: 28/06/26

1. AI Use Summary

Question	Student Response
Did you use any AI tool for this lab?	Yes
Estimated percentage of the work influenced by AI	10 %
Did you attach evidence of AI use where applicable?	Yes

2. Details of AI Use

Complete the table below for each AI tool used. Add more rows where necessary.

Name of AI Tool Used	Purpose for Using the Tool	Prompt or Instruction Given to the Tool	Part(s) of the Work Influenced by the Tool	How I Verified, Edited, or Improved the AI Output
Copilot	Debugging	I'm training the neural network from Part 1.3 with learning rates 0.001, 0.05, and 1.0. When using 1.0, I get warnings such as: RuntimeWarning: overflow encountered in square RuntimeWarning: overflow encountered in matmul Explain what these warnings mean, why they occur, and whether they	1.3	I verified the explanation by comparing it with my learning-rate graphs.

		indicate a bug in my code or a learning-rate problem.		
Copilot	Debugging	My NumPy neural network produces a matrix multiplication error during the forward pass. Help me trace the shape of every variable (x, W1, b1, z1, h, W2, y_hat) and identify where the dimension mismatch occurs. Explain how I can debug similar shape errors in future neural-network implementations	1.2	I verified the AI output by checking the shape of every variable in the forward pass and comparing them with the expected dimensions
Deepseek	Understanding the concepts	Act like my deep learning lecturer. Explain backpropagation in simple English using the neural network from Lab 3. Do not just give a definition. Show how errors move backward through the network, how gradients are calculated, how the weights are updated, and how this relates to loss.backward() in PyTorch.	Back propagation	Used other notes and external resources

3. Attachment of AI Output Evidence

Where required, attach evidence of AI use. Tick all that apply. Add more rows where necessary.

Evidence Type	Attached? (Yes, No, N/A)	File Name
AI-generated draft		
Prompt history		

Screenshot of AI interaction		Explain what these warnings mean, why they occur, and whether they indicate a bug in my code or a learning-rate problem. My NumPy neural network produces a matrix multiplication error during the forward pass. Help me trace the shape of every variable (x, W1, b1, z1, h, W2, y_hat) and identify where the dimension mismatch occurs. Explain how I can debug similar shape errors in future neural-network implementations
Exported AI conversation		
AI-generated code snippet		
Revised version showing student input		

4. Student Declaration

I declare that:

Declaration Statement	Response (Yes/No)
The submitted work is my own work.	Yes
Any use of AI tools has been clearly declared in this form.	Yes
The prompts, instructions, and AI-generated outputs have been disclosed where applicable.	Yes
I reviewed, tested, edited, and improved any AI-generated content before submission.	Yes
My AI usage does not exceed 25% of the entire work.	Yes
I understand that undeclared or excessive AI use may be treated as academic misconduct.	Yes

Student Signature:K.A.G _____

Date: 28/06/26 _____